UNLOCKING BEHAVIORAL INTELLIGENCE IN AIRLINE LOYALTY PROGRAMS

Churn Prediction, Customer Segmentation & a Data-Driven Retention Playbook

Technical Report

Prepared for: Marketing Leadership (CMO) and Finance Leadership (CFO)

Scope: 16,737 Canadian loyalty members | Activity data, 2017–2018

Prepared by: Business Analytics Team — Consulting & Analytics Project

Executive Summary

This report analyzes 16,737 Canadian airline loyalty members and 392,936 monthly flight-activity records (January 2017 – December 2018) to answer three questions for the marketing and operations teams: who is at risk of disengaging, what makes a member genuinely valuable, and what should the airline do about it — specifically and on a defined timeline.

What we did

- Defined churn to capture both formal cancellations (2,067 members) and "silent" disengagement — members who stop flying without ever formally cancelling (an additional 899 members), for a combined churn rate of 8.1% among members who have flown at least once.
- Engineered 29 behavioral, recency, trend, seasonal, and demographic features per member, built strictly from data available before the prediction window to avoid using future information.
- Segmented the base into four groups — Core Flyers, Drifting Loyalists, Silent Attritors, and Never Active — using behavioral clustering, then validated each segment against churn and CLV outcomes.
- Built and benchmarked three machine-learning models (XGBoost, LightGBM, Random Forest) and combined them into a single ensemble that flags churn risk with 0.998 ROC-AUC — in practical terms, the model correctly separates members who will disengage from those who will not in nearly every case.
- Translated model output into a rules-based retention playbook — every member receives a specific action, a trigger condition, and a timeline — and packaged it into an interactive Streamlit dashboard for non-technical use.

What we found

- 904 members (5.4% of the base) are at High or Critical risk of churning in the near term and require near-term outreach.
- **The most actionable group is "Drifting Loyalists" (3,467 members, 20.7% of the base):** they are still flying, but their churn rate (16.7%) is more than triple that of Core Flyers (4.6%). This is the segment where retention spend has the highest expected return, because these members are still reachable.
- **A counterintuitive finding:** the "Never Active" segment — 1,570 members who enrolled but never flew — carries the highest average CLV of any segment ($8,361) yet has a 60.6% churn rate. This indicates the current CLV metric overstates the realized value of dormant accounts and should not be used on its own to prioritize retention spend.
- **"Silent Attritors" (1,351 members) show an unusually high points-redemption rate** relative to other segments — consistent with a "redeem a large reward, then disengage" pattern that can serve as an early warning signal.

What we recommend

- Recommendation 1 – Targeted re-engagement for Drifting Loyalists: automated, trigger-based campaigns for the 537 members showing elevated churn risk within this segment, with a 25% reactivation target.

- Recommendation 2 – Last-mile win-back for high-value Silent Attritors: a tightly scoped, single-touch outreach to the 58 highest-value members in this segment, with an 8% win-back target and a hard stop on spend for non-responders.
- Recommendation 3 – Fix the CLV metric and activate the dormant base: review the CLV calculation to account for engagement, and launch a structured onboarding program for the 1,570 Never Active members, targeting a 20% first-flight conversion rate.

Each recommendation below specifies the trigger, the action, the owner-ready timeline, the expected impact, and the trade-off the leadership team should weigh before approving it.

1. Problem Framing & Scope

1.1 Business objective

Airlines invest heavily in loyalty programs to drive repeat business, but most programs remain reactive: marketing teams learn that a high-value member has stopped flying only after it is too late to intervene. The objective of this project is to make the airline's loyalty strategy proactive — to identify disengagement before it becomes permanent, to understand which members are genuinely worth retaining (beyond the headline CLV figure), and to give the marketing team a concrete, ready-to-use action for every member in the database.

1.2 Specific objectives and success criteria

- **Churn prediction without hindsight bias:** define what “churn” means for a loyalty program (beyond formal cancellation), and predict it using only information that would have been available at the time — success means the model generalizes to future months, not just explains the past.
- **Segmentation that goes beyond CLV:** produce member groups that are both statistically distinct and operationally meaningful — success means a marketing manager can look at a segment name and immediately know who is in it and what to do.
- **Actionable retention logic:** every recommendation must specify who receives it, when, in what form, and how success will be measured — generic advice (“offer points to at-risk members”) does not meet the bar.

1.3 Data sources, timeframe, and key metrics

The analysis draws on three linked data sources covering the full member lifecycle:

- **Customer Loyalty History** (16,737 members, one row per member): demographics (gender, education, salary, marital status), loyalty tier (Star / Nova / Aurora), Customer Lifetime Value (CLV), enrollment details, and formal cancellation records.
- **Customer Flight Activity** (392,936 rows, monthly): flights booked, distance flown, and points accumulated / redeemed for every member, January 2017 – December 2018.
- **Calendar dimension:** maps each month to its quarter and season, used to build seasonal travel features.

The key metrics used throughout this report are: CLV (the airline's existing value metric), total flights and distance flown, points earned vs. redeemed (“redemption rate”), recency (months since last flight), engagement ratio (share of months active), activity trend (recent vs. historical activity), and the model's churn probability and resulting risk tier.

2. Data Cleaning Decisions and Rationale

The flight activity file (392,936 rows) was complete — no missing values across any of its eight columns. The member-level file required four specific cleaning decisions, each documented below so that downstream users understand exactly what was changed and why.

2.1 Missing salary values (4,238 of 16,737 members, ~25%)

Salary was missing for roughly a quarter of members. Rather than drop these members or use a single global average (which would understate income differences across loyalty tiers), missing salaries were filled with the median salary for that member's loyalty card tier (Star: $73,135 · Nova: $73,327 · Aurora: $74,123). This preserves the tier-level income pattern while avoiding the loss of 25% of the member base from any analysis that uses salary.

2.2 Negative salary values (20 members)

Twenty members had negative salary values (e.g., –$57,297). All 20 belonged to the “2018 Promotion” enrollment cohort, indicating a sign error introduced during that batch's data entry rather than 20 genuinely negative incomes. These values were corrected by taking the absolute value, recovering the records without discarding them.

2.3 Cancellation date fields (14,670 of 16,737 members, ~88% null)

Cancellation Year and Cancellation Month are null for the vast majority of members — but this is a structural null, not missing data: it means the member has not cancelled. These fields were used to construct a binary “is_cancelled” flag (2,067 members = 1) rather than being imputed, which avoids accidentally treating “still enrolled” members as missing data.

2.4 Single-month activity records

A small number of members had only one month of recorded activity, which produces an undefined standard deviation for their monthly flight counts (a measure of consistency). These were set to 0, reflecting “no observed variability” rather than “unknown variability,” so they do not get incorrectly flagged as highly volatile.

2.5 Data quality limitation: a two-year window

The flight activity data covers only 2017–2018. This is enough to compare year-over-year and half-year activity (used to build the behavioral churn definition and the activity trend feature, see Section 3), but it is not enough to observe long-run, multi-year loyalty cycles. We mitigated this by using within-period comparisons (2017 vs. H1 2018, and H1 2018 vs. H2 2018) rather than relying on longer trends that the data cannot support. As more data accumulates, the churn definition and trend features should be re-validated against a longer history.

3. Modeling and Segmentation Logic

3.1 Defining churn (no churn label was provided)

The raw data contains a formal cancellation flag, but cancellation understates true disengagement — many members simply stop flying without ever formally leaving the program. Churn was therefore defined as the union of three signals:

1. **Formal cancellation** — the member's record shows a cancellation date (2,067 members).

2. **Year-over-year drop-off** — the member flew at least once in 2017 but recorded zero flights in all of 2018 (553 members).

3. **Within-2018 drop-off** — the member flew in the first half of 2018 but recorded zero flights in the second half (an additional 346 members).

Combining these three signals (with overlaps removed) gives 2,175 churned members. A separate group of 1,570 members never flew at all during the observation window. These members were deliberately excluded from the churn model: a member who was never engaged cannot "disengage," so labeling them as churned or not-churned would muddy the model's signal. Instead, this group is treated as its own segment ("Never Active") with its own recommendation (Section 5, Recommendation 3). Among the 15,167 members who flew at least once, 1,224 (8.1%) meet the combined churn definition — this is the population the churn model was trained and evaluated on.

3.2 Avoiding hindsight bias (no future information)

To ensure the model could be deployed in production — i.e., to predict the future, not just describe the past — the flight activity data was split into a feature window (January 2017 – June 2018, 291,530 rows) used to build every predictive feature, and a label window (July – December 2018, 101,406 rows) used only to determine whether a member churned. No information from the label window was used as a model input.

3.3 Feature engineering (29 features)

Every feature was built to answer a specific question about a member's relationship with the airline:

- **Flight behavior (volume):** total flights, distance, points earned and redeemed, active vs. inactive months, and the maximum, average, and consistency (standard deviation) of monthly flights — captures how much a member flies and how steady that pattern is.
- **Recency & engagement:** months since the member's last flight, the share of months they were active ("engagement ratio"), their redemption rate (points redeemed vs. earned), and average distance per flight — captures how recently and how meaningfully a member engaged.
- **Trend:** a comparison of first-half-2018 activity against a pro-rated 2017 baseline ("activity trend") — captures whether a member is accelerating or winding down, which is a stronger churn signal than activity level alone.
- **Seasonality:** flights taken in each season — captures whether a member is a seasonal traveler, useful for timing re-engagement campaigns.
- **Demographics:** CLV, salary, tenure, loyalty tier, gender, education, marital status, and promotional enrollment — provides context but, as discussed below, was secondary to behavior in predicting churn.

3.4 Segmentation: four member groups

Members who had flown at least once (15,167) were clustered using K-Means on eight behavioral and value features: engagement ratio, recency, total flights, CLV, activity trend, redemption rate, average distance per flight, and tenure. Cluster quality (silhouette score) was tested for K = 2 through 6; K = 3 was selected for business clarity — it produces groups that map cleanly onto distinct retention strategies, even though K = 2 scored marginally higher on the statistical metric alone. Combined with the separately-identified "Never Active" group (members who never flew), this produces four final segments:

Segment	Members	% of Base	Engagement	Recency (mo)	Avg CLV	Churn Rate
Core Flyers	10,349	61.8%	60.2%	0.8	$7,967	4.6%
Drifting Loyalists	3,467	20.7%	25.6%	1.3	$7,860	16.7%
Silent Attritors	1,351	8.1%	1.2%	17.6	$8,057	12.9%
Never Active	1,570	9.4%	—	—	$8,361	60.6%

Segment profiles. "Engagement" = share of months active; "Recency" = months since last flight.

- **Core Flyers (61.8% of base):** the engine of the loyalty program — highly engaged, recently active, and the lowest churn rate by a wide margin.
- **Drifting Loyalists (20.7% of base):** moderately engaged but with the highest churn rate among members who are still flying — the highest-leverage group for retention spend (Recommendation 1).
- **Silent Attritors (8.1% of base):** minimal recent activity (17.6 months since last flight on average) and an unusual redemption pattern (Section 4) — mostly past saving, with a small high-value subset worth one final attempt (Recommendation 2).
- **Never Active (9.4% of base):** enrolled but never flew — a CLV figure that looks attractive on paper but a 60.6% churn rate in practice (Recommendation 3).

3.5 Churn prediction model and validation

Three gradient-boosting and ensemble algorithms — XGBoost, LightGBM, and Random Forest — were each tuned via 5-fold cross-validation (with SMOTE used during training only, to ensure the model sees enough churn examples to learn from, since churners are a minority at 8.1%). The three tuned models were then combined into a soft-voting ensemble that averages their probability scores. Performance is summarized below, with each metric translated into plain terms:

- **ROC-AUC** (0.0–1.0, higher is better) — how reliably the model ranks members from most to least likely to churn. An ROC-AUC of 0.998 means that, almost without exception, the model correctly ranks an at-risk member above a safe member.
- **F1 Score** (0.0–1.0, higher is better) — a balance between catching true churners and avoiding false alarms. The ensemble's 0.93 means the model rarely misses a real churner and rarely cries wolf on a loyal member.
- **Precision** — of the members the model flags as "at risk," what share genuinely go on to churn. The ensemble's 0.925 means roughly 93 of every 100 flagged members are true churn risks — retention spend is not being wasted on the wrong people.
- **Recall** — of the members who actually churn, what share the model catches in advance. The ensemble's 0.936 means the model identifies about 94 of every 100 members who will churn, before they leave.

Model	ROC-AUC	F1 Score	Precision	Recall
XGBoost	0.979	0.803	0.829	0.779
LightGBM	0.997	0.943	0.966	0.921
Random Forest	0.999	0.963	0.949	0.977
Ensemble (final model)	0.998	0.930	0.925	0.936

Churn model performance. Higher is better on all four metrics (1.0 = perfect).

The ensemble was selected as the production model because it is the most robust to the quirks of any single algorithm. Each member's ensemble probability is converted into one of four risk tiers for operational use:

Risk Tier	Churn Probability	Members	Share of Base
Critical	≥ 70%	668	4.0%
High	50% – 70%	236	1.4%
Medium	30% – 50%	344	2.1%
Low	< 30%	15,489	92.5%

Risk tiers derived from the ensemble's churn probability. 904 members (Critical + High) are recommended for near-term outreach.

3.6 Interpretability for business stakeholders

The model's output is not used as a black-box score. Instead, each member's segment (Section 3.4) and risk tier (Section 3.5) are combined through a simple, transparent rules engine (Section 5 and Section 6) that converts "this member has a 74% churn probability" into "this member is a Silent Attritor with high CLV — call them within 7 days and offer 3,000 bonus points and 15% off." This means any marketing team member can audit why a given member received a given recommendation, without needing to interpret the underlying model.

A note on validation for ongoing use: the performance figures above reflect cross-validated tuning on the available 2017–2018 history. Before this model drives live spend at scale, we recommend a short out-of-time validation — scoring the most recent 1–2 months of new activity once it becomes available — to confirm performance holds as travel patterns evolve season to season.

4. Findings and Insights

4.1 Behavior, not background, drives churn

Across the feature set, recency, engagement ratio, and activity trend were the dominant signals separating churners from active members — far more than demographic attributes like education, marital status, or salary. In plain terms: what a member does in the months before they leave matters far more than who they are. This has a direct operational implication — the marketing team does not need rich demographic profiling to act on this model; recent flight activity is enough.

4.2 The base is healthier than it looks — but one segment needs attention now

Nearly two-thirds of the base (Core Flyers) churns at under 5% — a strong foundation. The concern is concentrated: Drifting Loyalists churn at more than three times that rate (16.7%) while still being reachable (unlike Silent Attritors, who have been gone for over a year on average). Of the 904 members flagged High or Critical risk, the majority sit within Drifting Loyalists and Silent Attritors — confirming that risk is concentrated, not spread evenly across the base.

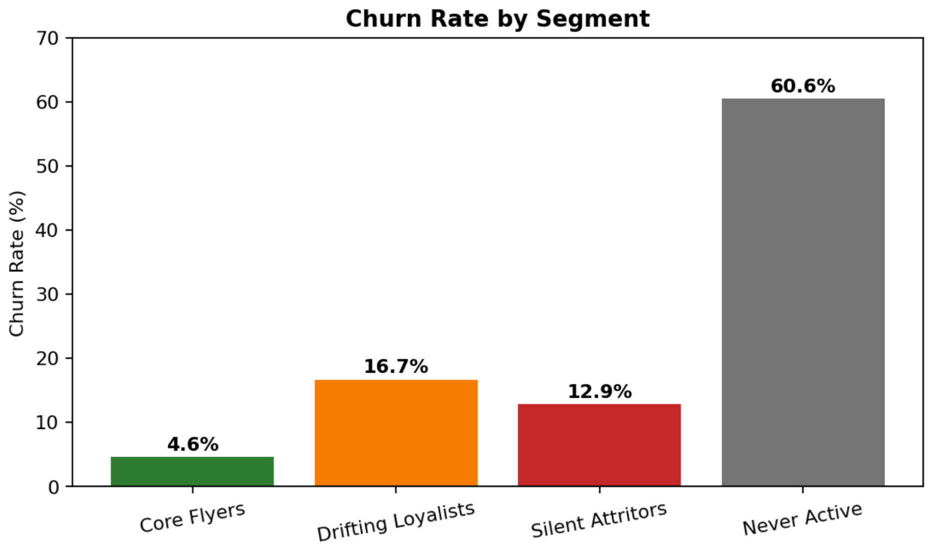

Figure 1. Churn rate by segment. Drifting Loyalists churn at more than 3x the rate of Core Flyers while remaining engaged enough to be re-activated.

4.3 Surprising finding: the CLV metric overstates dormant value

The single most counterintuitive result of this analysis is that the “Never Active” segment — members who enrolled but have never taken a flight — carries the highest average CLV of any segment ($8,361), yet has a 60.6% churn rate, the highest in the program.

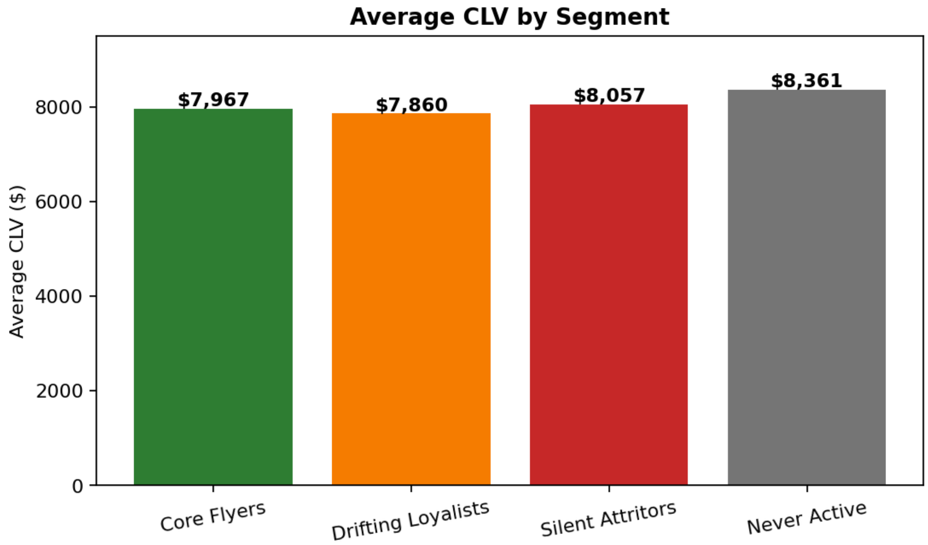

Figure 2. Average CLV by segment. CLV is nearly flat across segments — it does not distinguish a highly engaged Core Flyer from a member who has never boarded a flight.

This matters because if CLV is used (even informally) to prioritize where marketing spend goes, it currently sends exactly the wrong signal for this group: a high “value” figure attached to members who are, behaviorally, the least connected to the airline. Recommendation 3 addresses this directly.

4.4 Surprising finding: a “redeem and run” pattern in Silent Attritors

Silent Attritors show a redemption rate (points redeemed relative to points earned) far above any other segment — roughly two and a half times their points balance on average, compared to roughly 2% for Core Flyers and Drifting Loyalists. The most plausible explanation is a one-time, large redemption (e.g., a reward flight or major purchase) shortly before the member stops engaging. This pattern is a useful early-warning signal in its own right: a sudden, large redemption combined with declining recent activity may flag a member who is about to become a Silent Attritor — worth monitoring as the model is refreshed over time.

4.5 Risk is concentrated, not diffuse

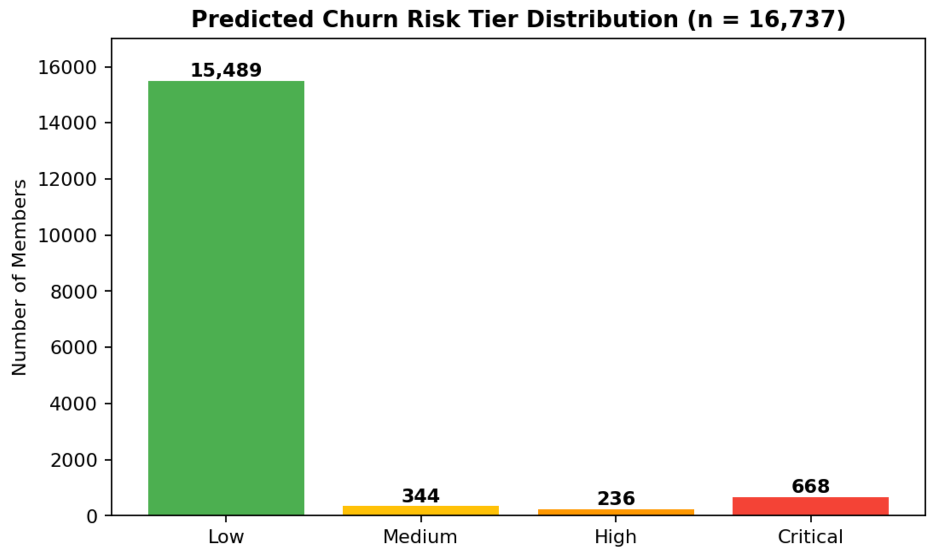

Figure 3. Predicted churn risk tier distribution. 92.5% of members are Low risk; the remaining 7.5% (1,248 members across Medium/High/Critical) are where retention effort should be focused, with 904 in High/Critical requiring near-term action.

5. Business Recommendations

Each recommendation below is scoped to a specific, named group of members, with a defined trigger, a concrete action, an owner-ready timeline, the metric used to measure success, the expected impact, and the trade-off the leadership team should weigh.

Recommendation 1: Targeted Re-engagement for Drifting Loyalists

- **Who:** 537 members within the Drifting Loyalists segment whose churn probability exceeds 40% and who have had no flights in the past 3+ months (out of 3,467 members in this segment).
- **Trigger:** automated, refreshed monthly — churn probability > 40% AND no flights in 3+ consecutive months.
- **Action:** 2x points on the member's next booking, paired with a personalized email highlighting routes the member has flown before. The remaining, lower-risk members of this segment receive a lighter-touch seasonal nudge (1.5x points for advance bookings, sent each May 1 and November 1).
- **Timeline:** outreach within 30 days of the inactivity signal firing; campaign performance reviewed monthly alongside the model refresh.

- **Owner & data readiness:** marketing automation platform capable of trigger-based email and points-offer campaigns; requires monthly refresh of churn scores from the model (already produced as part of this deliverable).
- **Expected impact:** this segment alone represents roughly $27 million in CLV (3,467 members × ~$7,860 average). A 25% reactivation rate among the 537 targeted members — the benchmark used in the underlying playbook — would directly protect a meaningful share of that CLV from attrition. Tracking metric: % of targeted members flying within 60 days of outreach.
- **Trade-off:** 2x points campaigns carry a real cost and, if extended beyond the flagged 537 members, would erode margin without a corresponding retention benefit. The recommendation is deliberately scoped to the highest-risk subset; the broader segment receives only the lower-cost seasonal nudge.

Recommendation 2: Last-Mile Win-Back for High-Value Silent Attritors

- **Who:** 58 members within the Silent Attritors segment with CLV above $7,000 and a churn probability above 70% (the “Critical” intersection of this segment). The remaining 1,293 lower-value Silent Attritors are routed to a low-cost “Graceful Exit.”
- **Trigger:** recency greater than 24 months AND churn probability above 70% AND CLV above $7,000.
- **Action:** a single, differentiated outreach attempt — Aurora and Nova tier members receive a direct phone call plus 3,000 bonus points and 15% off their next booking; Star tier members receive an email with 1,500 bonus points and 10% off. Members not responding within 30 days are archived from active marketing (no further spend).
- **Timeline:** one attempt within 7 days of identification; archive after 30 days with no response.
- **Owner & data readiness:** CRM with phone-outreach capability for top-tier members; a hard-coded archive rule to stop spend automatically after the 30-day window.
- **Expected impact:** even at a conservative 8% win-back rate (the playbook's benchmark), this recovers approximately 4–5 high-value members at an average CLV near $8,000 — a meaningful return relative to the incentive cost of a single call and points award. Critically, the remaining 1,293 lower-value Silent Attritors are moved to a single-send “graceful exit” (a final 500-point email) and then removed from active campaigns, which stops marketing spend on a group with a low probability of return.
- **Trade-off:** direct phone outreach is the most expensive channel per contact in this playbook; it is therefore capped at 58 members. Expanding it to the full Silent Attritors segment (1,351 members) would multiply the cost without a proportional increase in win-backs, given this segment's 17.6-month average recency.

Recommendation 3: Correct the CLV Metric and Activate the Dormant Base

- **Who:** 1,570 “Never Active” members (enrolled but with zero recorded flights), plus — for the CLV component — the airline's broader CLV methodology as applied across all segments.
- **Trigger (CFO action):** the finding in Section 4.3 — CLV is currently flat across segments and does not reflect engagement, meaning a dormant member and a Core Flyer can carry similar headline value.
- **Action (CFO-facing):** commission a focused review, with Finance and the Data team, of the CLV calculation to incorporate an engagement or activity multiplier — so that

reported portfolio value reflects realized engagement, not just a static projection. This directly improves the accuracy of Recommendations 1 and 2, which both reference CLV.

- **Action (CMO-facing):** launch a "Welcome Activation" program for the 1,570 Never Active members — a 500-point bonus on their first booking, supported by an onboarding email series sent at months 1, 2, and 3 after enrollment.
- **Timeline:** CLV methodology review — 1–2 working sprints with Finance; Welcome Activation — begins immediately for existing Never Active members and is built into the standard onboarding flow for all future enrollees. Tracking metric: first-flight conversion rate within 90 days of the campaign.
- **Owner & data readiness:** onboarding email automation (marketing); CLV methodology review (finance + data team).
- **Expected impact:** the playbook targets a 20% first-flight conversion rate — roughly 314 of the 1,570 Never Active members beginning to fly within the tracking window, moving them toward the Core Flyer profile over time. Separately, correcting the CLV methodology improves the accuracy of every value-based decision the airline makes going forward, including the prioritization logic in Recommendations 1 and 2.
- **Trade-off:** activation incentives (500 points per member) are a guaranteed cost against an uncertain 20% conversion outcome — 80% of the cost will go to members who do not convert. Separately, a corrected CLV methodology may produce a one-time reduction in the airline's reported "total portfolio CLV" figure; this should be flagged to Finance in advance so it is understood as a measurement correction, not a real loss of value.

6. Deliverables and Reproducibility

6.1 What was built

- **Analysis notebook:** a single, end-to-end notebook covering data loading, the four cleaning steps in Section 2, feature engineering, segmentation, the three-model churn ensemble, risk tiering, and the recommendation engine described in Section 5.
- **Scored member file (master_final.csv):** all 16,737 members with their full feature set, segment, churn probability, risk tier, and assigned retention action — 47 columns in total.
- **At-risk file (at_risk_customers.csv):** the 904 High/Critical risk members, sorted by churn probability, ready for the marketing team to action immediately.
- **Interactive dashboard (Streamlit app):** a four-tab interface — Overview (segment mix, churn-risk distribution, CLV vs. engagement), At-Risk Customers (filterable, downloadable action list), Segments (deep-dive profiles per segment), and Retention Playbook (the trigger / action / timeline / KPI for each segment, plus an individual member lookup). Sidebar filters allow the marketing team to slice by segment, risk tier, card tier, and CLV range without any technical setup.

Follow the link below to the dashboard-

https://airline-loyalty-dashboard.streamlit.app/

6.2 Recommendation engine — full action summary

The table below shows how the 16,737 members are distributed across the segment × priority × action combinations described in Section 5 and embedded in the dashboard's Retention Playbook tab.

Segment	Priority	Action	Members
Core Flyers	High	Tier Protection Alert	188
Core Flyers	Low	Loyalty Deepening	10,161
Drifting Loyalists	Medium	Re-engagement Campaign	537
Drifting Loyalists	Low	Seasonal Nudge	2,930
Silent Attritors	Critical	Last-Mile Win-Back	58
Silent Attritors	Low	Graceful Exit	1,293
Never Active	Low	Welcome Activation	1,570

Full distribution of members across retention actions. The 904 High/Critical/Medium-priority members above are the near-term focus of Recommendations 1 and 2.

6.3 Key assumptions for non-technical readers

- “Churn” = formal cancellation OR a member who stopped flying for an extended period (Section 3.1). Members who never flew are tracked separately and are not counted as “churned” from a relationship they never entered.
- All predictive features use only data available before the prediction window (Section 3.2) — the model is built to work on new, incoming data, not just to explain the past.
- Segments (Core Flyers, Drifting Loyalists, Silent Attritors, Never Active) are derived from behavior and value, not assumed — every member's segment can be traced back to their engagement ratio, recency, CLV, activity trend, redemption rate, distance per flight, and tenure (Section 3.4).
- Every recommendation in Section 5 is reproducible directly from the scored member file — the “who,” “trigger,” and “action” columns in master_final.csv map one-to-one with the logic described above.

Appendix

A1. Loyalty Base by Segment

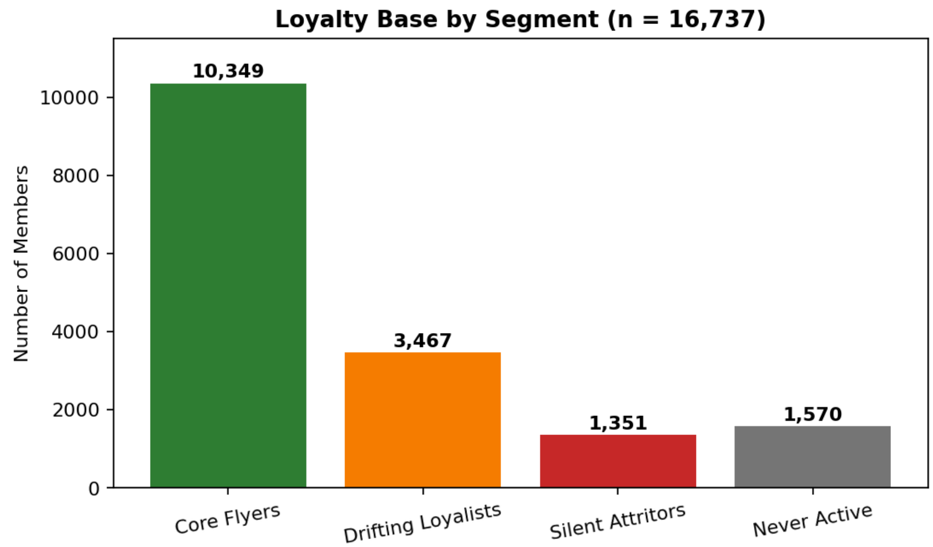

Figure A1. Distribution of all 16,737 members across the four segments.

A2. Churn Model Performance Comparison

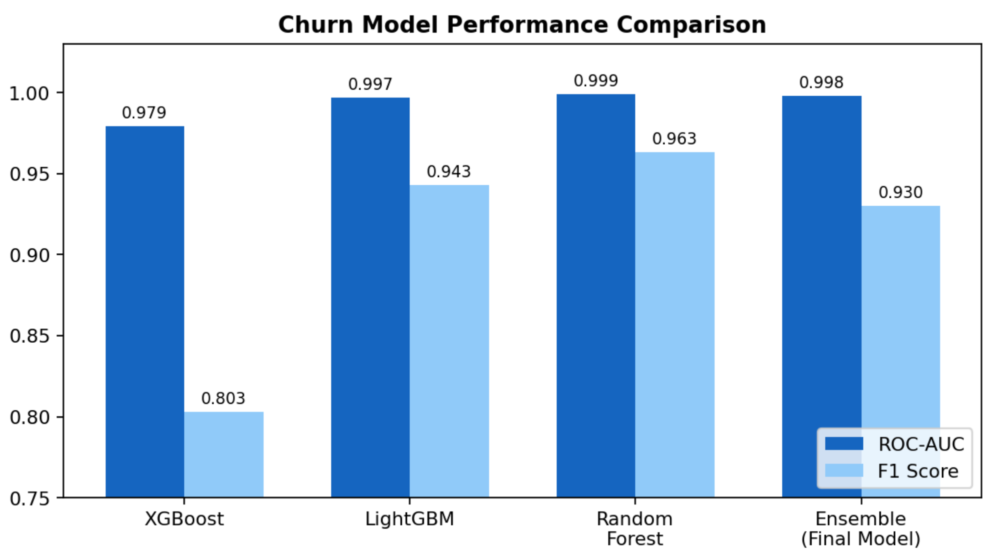

Figure A2. ROC-AUC (how reliably the model ranks at-risk members above safe ones) and F1 score (the balance between catching churners and avoiding false alarms) for each individual model and the final ensemble. The ensemble was selected for production because it is more robust than any single model across both metrics.

A3. Data Dictionary and Methodology Notes

This table summarizes the fields used throughout the analysis, grouped by source and purpose, for technical readers who wish to trace any finding back to its underlying data.

Field (group)	Description	Why it matters
Loyalty Number, Gender, Education, Salary, Marital Status, Loyalty Card, CLV, Enrollment Type/Year/Month, Cancellation Year/Month	Member demographics, tier, and enrollment / cancellation history (source: Customer Loyalty History)	Baseline profile and the only source of a formal “cancelled” signal
Year, Month, Total Flights, Distance, Points Accumulated, Points Redeemed, Dollar Cost Points Redeemed	Monthly flight activity per member, Jan 2017 – Dec 2018 (source: Customer Flight Activity)	Core behavioral signal; basis for all engineered engagement and recency features
tenure_months, card_rank	Months since enrollment; numeric rank of loyalty tier (Star/Nova/Aurora)	Captures relationship length and current tier
is_cancelled, behavioral_churn, churn, never_flew	Engineered churn flags (see Section 3)	Defines the prediction target and excludes never-active members from the churn model
total_flights, total_distance, total_pts_earned/redeemed, active_months, zero_months, max/avg/std_flights_month	Aggregated flight activity, Jan 2017 – Jun 2018 (feature window only)	Volume and consistency of engagement
recency_months, engagement_ratio, redemption_rate, avg_dist_per_flight, activity_trend, flights_2017, flights_h1_2018	Recency, engagement intensity, redemption behavior, and growth/decline trend	Strongest predictive signals – capture momentum, not just history
fall, spring, summer, winter	Flights taken in each season (feature window)	Surfaces seasonal travel patterns for targeted campaign timing
churn_prob, risk_tier, segment, priority, action, what_to_do, when_to_act	Model outputs and recommendation engine outputs	Drives the dashboard and the retention playbook

A4. Reference to source materials

Full code, model objects, hyperparameter search results, and the Streamlit dashboard source are available in the project notebook and accompanying app.py file referenced in Section 6.1. The notebook is organized to mirror the structure of this report: data overview and cleaning, feature engineering, segmentation, churn modeling and benchmarking, risk tiering, the recommendation engine, and dashboard export.